Zhaofu (Jeff) Chen

Irvine, CA

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Materials research | Conducting polymers | ENZ optics

Education

University of California, Irvine Expected Jun 2027

B.S. Chemical & Biomolecular Engineering; Minor in Materials Science and Engineering

Henry Samueli School of Engineering

Publications & Manuscripts

Peer-reviewed publications

1. **3D nanoscale design of oxide morphology, anisotropy, and chemical heterogeneity by glancing angle physical vapor deposition.**
   Nanoscale (2026). doi: 10.1039/d6nr01275b. Fourth author.
2. Guo, L., **Chen, Z.**, et al. "Advances and outlook in modified graphene oxide (GO)/epoxy composites for mechanical applications." *Applied Nanoscience* **13**, 3273–3287 (2023).

Submitted manuscripts

3. **Enhanced Third-Harmonic Generation in PEDOT:PSS Polymer Films Enabled by Femtosecond Laser Irradiation.**
   Fourth author. Submitted to *Nature Materials*.
4. **Solid-state synthesis & characterization of indium-oxide-based medium-entropy oxides.**
   Co-first author, equal contribution. Submitted to *Journal of the American Ceramic Society*; ChemRxiv preprint (2025).

Manuscripts in preparation

5. **Tunable Near-Infrared Epsilon-Near-Zero Response in PEDOT:PSS Films via Thermal and Ethylene-Glycol Processing.**
   Co-first author, equal contribution. In preparation for submission to *Advanced Optical Materials*; working title.
6. **Tunable Conducting Polymers for Nonlinear and Epsilon-Near-Zero Optics: A Review.**
   In preparation for submission to *Laser & Photonics Reviews*.
7. **Thermal-Stability Design Strategies for Non-Fullerene Organic Photovoltaics.**
   Review manuscript in preparation; working title.
8. **Amorphous high-entropy oxides for CO_2 electroreduction.**
   Co-first author, equal contribution. In preparation; planned submission to *Advanced Materials*.

Conferences & Presentations

- **APS Far West Section Meeting** 2026
  Presenting author; poster contribution. Chen, Z., Yan, J., Anopchenko, A., Chao, A., and Lee, H. W. "Tailoring Near-Infrared Epsilon-Near-Zero Optical Resonances in Conducting Polymer Thin Films via Controlled Processing."
- **UCI UROP Symposium & IRT Project Symposium** Jun 2025; Jun 2026
  Research presentations at both symposia in 2025 and 2026.
- **UCI Engineering Council Research Fair** Nov 2025
  Poster presentation.
- **Optica Advanced Photonics Congress** — Co-author on conference submission.
- **SPIE Nanoscience + Engineering** — Co-author on conference submission: "Enhanced Third-Harmonic Generation in PEDOT:PSS Polymer Films Enabled by Femtosecond Laser Irradiation."

Research Experience

Lee Nano-Optics Lab, University of California, Irvine 2025–Present

Undergraduate Researcher | Advisor: Prof. Ho Wai (Howard) Lee

Tunable Near-Infrared ENZ in PEDOT:PSS Thin Films

- Fabricate three-layer PEDOT:PSS films and tune near-infrared epsilon-near-zero (ENZ) response through thermal annealing and ethylene-glycol post-treatment.
- Fit **spectroscopic ellipsometry** data to extract **film thickness** and complex permittivity; obtained fitted ENZ wavelengths of **971.5 nm** (pristine), **953.7 nm** (annealed), and **875.5 nm** (EG-treated), a **96.0 nm blue shift** versus pristine.
- Characterize THG with **Leica SP8 microscopy**; **co-first author** on an ENZ manuscript in preparation for submission to *Advanced Optical Materials*.

Femtosecond Laser Writing & Third-Harmonic Generation in PEDOT:PSS

- Fabricate and characterize PEDOT:PSS films with controlled formulations and thermal treatment to investigate how femtosecond laser processing changes their nonlinear optical response.
- Measure **third-harmonic generation (THG)** using the **Leica SP8 nonlinear optical microscope** at the Beckman Laser Institute; use ellipsometry to characterize film thickness and optical constants.
- Contributed film fabrication and characterization to a **fourth-author manuscript submitted to *Nature Materials***.

Bias-Dependent Nanoscale Transport in PEDOT:PSS via AFM/CAFM

- Work with Christopher Gonzalez on **AFM/conductive AFM** studies of PEDOT:PSS under different applied biases, relating surface morphology to local current and I–V response.
- Contribute sample preparation and substrate design for measurements on conductive supports; assess how electrode interfaces, wetting, and film thickness affect interpretation of local transport.
- Analyzed AFM data in **Gwyddion**, applying leveling, scan-line corrections, and boundary-based step-height extraction to quantify film thickness and reduce tilt artifacts.

ITO/AZO ENZ Thin Films for Field-Resolved Mid-IR Sampling *(2026–Present)*

- Annealed ITO and fabricated ALD-grown AZO films to tune carrier response and optical dispersion toward a target ENZ wavelength near 2300 nm.
- Extracted optical constants and dielectric functions by ellipsometry to assess films for nJ-level, MHz few-cycle mid-IR sampling in a UC Irvine–Max Planck Institute collaboration.

Conducting-Polymer Review for Nonlinear & ENZ Optics

- Synthesized processing–structure–optical-property relationships and contributed figures, modeling workflows, and revisions for a review in preparation for *Laser & Photonics Reviews*.

Nguyen Lab, University of California, Santa Barbara Summer 2026

Summer Researcher | Advisor: Prof. Thuc-Quyen Nguyen

Thermal-Stability Design Strategies for Non-Fullerene Organic Photovoltaics

- Compared crosslinker and thermoset-matrix strategies for suppressing thermally driven morphology changes while preserving initial power-conversion efficiency and charge transport.
- Developed a design framework linking network formation to the location of thermal failure and retention of device function; preparing a review manuscript on these strategies.
- Proposed a controlled PM6:Y6 study with crosslinker-loading series, UV-only/nonactivated controls, and 85 °C aging, pairing device measurements with morphology, transport, and chemical-integrity readouts.

Research Experience (continued)

Bowman Lab, University of California, Irvine 2024–Present

Undergraduate Researcher | Advisor: Prof. William Bowman

Indium-Oxide-Based Medium-Entropy Oxides

- Investigated composition–processing–phase stability relationships through solid-state synthesis and XRD phase identification, indexing, and quantitative analysis; constructed processing–phase maps to guide sintering-window selection.
- Built Python/MATLAB workflows for batch comparison and publication figures; **co-first author** on a ChemRxiv preprint and manuscript submitted to *Journal of the American Ceramic Society*.

Amorphous High-Entropy Oxides for CO_2 Electroreduction

- Fitted EIS equivalent circuits across oxide compositions and developed four-probe/temperature-ramping workflows, resolving contact instability to support reliable electrical characterization.
- Integrated XRD, XPS oxygen-vacancy analysis, SEM–EDS, and PLD workflows to relate structure and chemistry to electrochemical response; led proposal development and manuscript preparation as a **co-first author**.

Complex Concentrated Oxide Thin Films for Memristors *(2025–Present)*

- Led proposal and experimental-roadmap development for oxygen-vacancy control in $InSnZnO_x$-based devices; secured a **$3,000 CALIT2 UROP IRT grant as project lead**.
- Automated I–V analysis, cycle statistics, threshold-voltage extraction, and resistance-state comparisons; applied electron-microscopy and spectroscopy workflows to investigate structure–switching relationships.

YSZ Thin Films & Glancing-Angle Deposition Architectures

- Analyzed XRD peaks and preferred orientation in PLD-grown YSZ films supporting glancing-angle-deposited architectures; automated batch comparisons and contributed to a **fourth-author publication in *Nanoscale* (2026)**.

Honors, Fellowships & Research Grants

- **Summer QUROP Award**, Eddleman Quantum Institute, UCI 2026 | $3,000
- **MSA Undergraduate Research Scholarship** 2026 | $3,000
- **CALIT2 UROP IRT Research Grant**, Project Lead 2025–2026 | $3,000
- **CALIT2 IRT Research Grant** 2024–2025 | $3,000

UROP Researcher (2024–2025); UROP Project Lead (2025–2026); CALIT2 IRT Research Fellow (2024–Present).

Technical Skills

Thin-film fabrication: PLD; ALD AZO; ITO annealing; PEDOT:PSS formulation; spin/drop coating; EG/MLLC processing; GOPS crosslinking; femtosecond laser writing; acid treatment.
Characterization: THG microscopy (Leica SP8); spectroscopic ellipsometry; Raman; optical transmission/absorption; XRD; XPS; SEM–EDS; AFM/CAFM; TEM/STEM-HAADF; four-probe and I–V measurements; EIS.
Analysis & software: Optical-constant extraction; EIS fitting; microscopy image analysis; regression and curve fitting; Python; MATLAB; Origin; ImageJ; Gwyddion; Adobe Illustrator; Excel.

Industry Experience

GoldenKeys High-tech Materials Co., Ltd. Jul–Aug 2024

Technical Control & Quality Intern

Automated Opentrons FLEX testing and MATLAB/Python data processing for oxidation-catalyst development; implemented data schemas and quality checks to improve traceability.